

AYUSH DAS

Machine Learning Engineer | LLMs & RAG | NLP & Deep Learning
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SUMMARY

Machine Learning Engineer specializing in LLM systems, RAG pipelines, and multi-agent orchestration. Delivered six production-grade AI projects — a citation-grounded medical RAG assistant, an autonomous 5-agent research pipeline, and an ML carbon-footprint estimator among them — each with a live demo and public codebase, plus a documentation fix merged into OpenCV (80k★ stars). Hands-on with LangGraph, ChromaDB, FAISS, DeBERTa-v3, and FastAPI, with a strong foundation in probability, statistics, and retrieval system design.

TECHNICAL SKILLS

LLMs & Agents: LangChain, LangGraph, RAG, Agentic AI Systems, Prompt Engineering, Transformers, Hugging Face, DeBERTa-v3, NLI, LangSmith, OpenAI API

Vector & Retrieval: FAISS, ChromaDB, sentence-transformers, Dense Retrieval, Semantic Search, Cosine Similarity, Embedding Pipelines

ML & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, XGBoost, Random Forest, Fine-Tuning, Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation

Deployment & MLOps: FastAPI, Django, REST APIs, Streamlit, GitHub Actions (CI/CD), Docker, Vercel, Hugging Face Spaces, Server-Sent Events, pytest, Git, GitHub

Data & Foundations: Pandas, NumPy, SQL, Probability, Statistics, Linear Algebra, Data Structures & Algorithms

Languages: Python (primary), C++, C, SQL, JavaScript

PROJECTS

Healthcare Knowledge Navigator — Citation-Grounded Medical RAG Assistant — [GitHub](#) · [Live Demo](#)

Jun 2026

- Built a zero-hallucination clinical RAG assistant grounding gpt-4o-mini in top-4 ChromaDB retrievals over OpenAI text-embedding-3-small, with inline [n] citations and an explicit refusal path when evidence is insufficient.
- Architected a single-origin FastAPI service serving UI and API together — eliminating CORS "unable to fetch" failures — with human-readable errors and an /api/health readiness probe.
- Engineered auto topic-discovery from indexed metadata and a 3-tier confidence meter (HIGH / MEDIUM / INSUFFICIENT); deployed to Vercel on a serverless-patched ChromaDB store.

C4Future — ML-Powered Carbon Footprint Generator — [GitHub](#) · [Live Demo](#)

Jun 2026

- Developed an AI sustainability tool predicting a product's CO2 footprint with a Random Forest Regressor (scikit-learn), splitting each estimate across 3 emission categories — materials, manufacturing, transport.
- Modeled the emission-factor dataset and training pipeline (Pandas/NumPy) from IPCC and logistics standards for stable predictions across product categories.
- Deployed an interactive Django + Chart.js app with carbon-offset strategies and real-world equivalencies (car-km, phone charges) on Hugging Face Spaces.

AI Research Assistant — Autonomous Multi-Agent RAG Pipeline — [GitHub](#) · [Live Demo](#)

Jun 2026

- Automated end-to-end research synthesis with a 5-agent LangGraph StateGraph (Planner → Searcher → Reader → Critic → Writer), using conditional routing and a self-critique retry loop to emit a sourced report per run.
- Designed a dual-layer memory — typed graph state for session context and persistent ChromaDB (all-MiniLM-L6-v2, top-3 recall) — improving relevance on repeated and related queries.
- Streamed live agent execution to the browser via FastAPI + Server-Sent Events with LangSmith tracing and graceful degradation (JSON retries, failed-URL skips, memory fallbacks).

Legal Clause Conflict Resolver — Explainable Legal NLP — [GitHub](#) · [Live Demo](#)

May 2026

- Fine-tuned DeBERTa-v3-large (NLI) to flag contradictory legal clause pairs, reaching >85% validation accuracy and ~15% fewer false positives vs. a zero-shot baseline.

- Integrated FAISS + sentence-transformers to pre-align related clauses, cutting redundant model calls on large document sets.
- Implemented cross-attention attribution heatmaps and a severity scorer to rank highest-risk conflicts first, exposed via a FastAPI backend and analytics dashboard.

Library Circulation Intelligence — Clustering + Semantic Retrieval (RAG) — [GitHub](#) · [Live Demo](#)

May 2026

- Engineered an unsupervised clustering pipeline over 119K+ Seattle Public Library records — 8 behavioral features, optimal k=6 by silhouette analysis (0.42, Davies-Bouldin 0.82) — yielding reproducible patron segments.
- Layered semantic retrieval (MiniLM embeddings + TF-IDF/LSA hybrid ranking) for cosine and book-similarity search through a FastAPI backend.
- Shipped a Next.js + Vercel frontend with fully client-side query embedding, packaged behind a tested CI pipeline for reproducible re-runs.

YT Genius — RAG Chatbot over YouTube Transcripts — [GitHub](#) · [Live Demo](#)

Dec 2025

- Built a hallucination-resistant RAG chatbot over YouTube transcripts — 500-char chunks (80-char overlap) and top-5 FAISS IndexFlatIP (cosine) retrieval before generation.
- Implemented the retrieval stack with OpenAI text-embedding-3-small + LangChain, grounding gpt-4o-mini strictly in retrieved context.
- Delivered a FastAPI + React/Vite app with token-by-token streaming and transcript ingestion for any YouTube URL.

OPEN-SOURCE CONTRIBUTION

OpenCV — Computer-Vision Library (80k+ ★) — [Pull Request #29299](#)

Merged · v4.14.0

- Authored and merged a documentation pull request fixing errors across 9 tutorial files (DNN, calib3d, ArUco, JavaScript bindings); accepted by a core maintainer into the 4.x branch (release 4.14.0).
- Navigated the full open-source workflow — fork, feature branch, CI checks, and maintainer code review — to land a change in a repository used by millions of developers worldwide.

EXPERIENCE

Web Development Intern — VanillaCart | Certificate

Sep 2025 – Nov 2025

- Rebuilt and launched a fully responsive production website — restructured page hierarchy and applied semantic HTML and on-page SEO (metadata, asset optimization) to improve crawlability and page-load performance.
- Partnered with the engineering team to coordinate deployment and QA, shipping cross-browser, mobile-responsive pages to production.

CERTIFICATIONS & CONTINUOUS LEARNING

RAG & LLMs: [Coursera – Introduction to Retrieval-Augmented Generation \(Jun 2026\)](#) · [IBM – ChatGPT-4 Prompt Engineering](#) · [Google – Prompt Engineering Essentials](#) · [Udemy – Generative AI & Tools](#)

ML Foundations: [DeepLearning.AI – Supervised Machine Learning \(Jun 2026\)](#) · [Microsoft – Foundations of AI & Machine Learning \(Jun 2026\)](#) · [CipherSchools – Machine Learning with Data Science](#) · [IBM – Python for Data Science & AI](#)

EDUCATION

B.Tech, Computer Science & Engineering

Aug 2023 – Present

Lovely Professional University, Punjab, India — CGPA: 8.08 / 10

Senior Secondary (Class XII)

Aug 2022

Hindi Higher Secondary School, Tripura, India — Percentage: 88%

Secondary (Class X)

Aug 2020

Holy Cross School, Tripura, India — Percentage: 94%